

UNIGE - Mobile Robots Exam - 2020-2021 - session 2

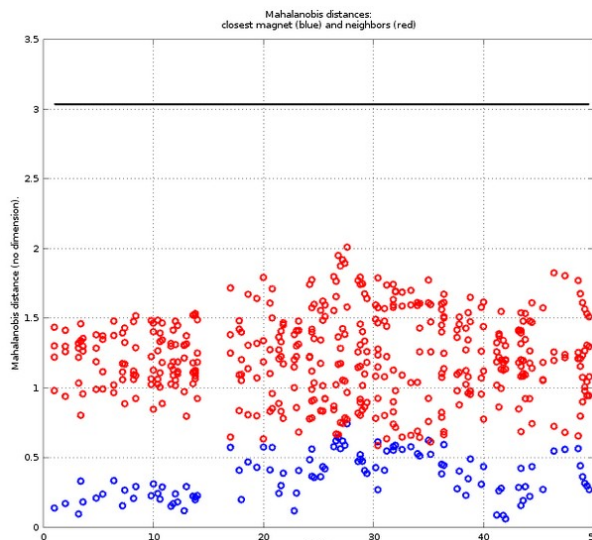
Duration: 2 h

Important warning: The present exam subject contains a number of questions for which you have to write text rather than equations. **Pay attention to your handwriting!** I should not have to guess. When not legible, there will be no benefit of the doubt: I shall consider the answer as wrong. Likewise, **make sure you use proper vocabulary.**

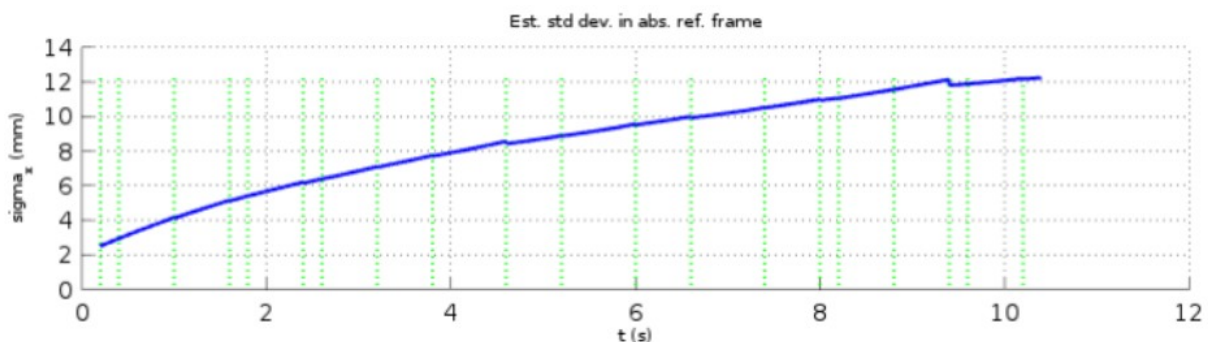
Exercise 1 - Lab related questions (30 min)

To solve the following questions, you have significantly more time than you need to write the answers. In the grading, **emphasis will be on clarity and choice of the most convincing arguments.**

1.1. The graph below shows the Mahalanobis distances obtained in an execution of the Kalman filter of the lab. Assuming the problem is due to an overestimation of either Q_γ or σ_{tuning} , explain why it has to be Q_γ . Use the equations of the Kalman filter to support your reasoning. Indicative length: 10 lines.



1.2. In an experiment of the lab where only the y measurement is used, the estimated standard deviation of the x position behaves in the following way, and so does the estimated standard deviation of the y position (not shown in the figure).



Answer the following questions:

1.2.a. What is happening to the overall localization uncertainty?

Is this behavior satisfactory? (justify in a maximum of two lines of text).

1.2.b. What do the green dotted lines represent?

Does the robot use measurements? (two lines of text maximum).

1.2.c. By checking multiple data sets, you realize that this happens only on straight line motions. What would you intuitively conclude? What theoretical aspect of the state estimator do you recommend to check? (five lines of text).

1.3. Explain why, in the context of the lab, it is fundamental to properly determine all variances (P_{init} , Q_γ , Q_α) and the Mahalanobis distance threshold **before** tuning the σ_{tuning} parameter.

(10 lines of text maximum).

1.4. Explain why σ_{tuning} is tuned by trial and error.

(10 lines of text maximum).

1.5. In the case when the localization system uses only the measurement of the y coordinate of the magnet in the robot frame, the equations are the following:

$$X = [x, y, \theta]^t \quad U = [v, \omega]^t$$

$$\dot{X} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} v \cos \theta \\ v \sin \theta \\ \omega \end{bmatrix} = f(X, U) \text{ (that's the continuous form, but it's not relevant here).}$$

$$Y = g(X) = \cos \theta ({}^0y_{mag} - y) - \sin \theta ({}^0x_{mag} - x)$$

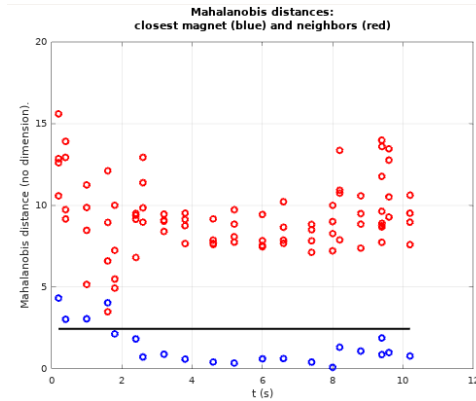
Where ${}^0x_{mag}$ and ${}^0y_{mag}$ are the coordinates of the detected magnet in R_0 .

In this context, there is no “wrong beacon”: when a magnet is detected, it can only be a magnet of the regular grid used by the localization system. Using the χ^2 table below, how do you define the threshold for the Mahalanobis distance in the above situation? Explain very shortly.

	$P(X \leq x)$							
	0.010	0.025	0.050	0.100	0.900	0.950	0.975	0.990
r	$\chi^2_{0.99}(r)$	$\chi^2_{0.975}(r)$	$\chi^2_{0.95}(r)$	$\chi^2_{0.90}(r)$	$\chi^2_{0.10}(r)$	$\chi^2_{0.05}(r)$	$\chi^2_{0.025}(r)$	$\chi^2_{0.01}(r)$
1	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.34
4	0.297	0.484	0.711	1.064	7.779	9.488	11.14	13.28
5	0.554	0.831	1.145	1.610	9.236	11.07	12.83	15.09

χ^2 table extract.

1.6. The figure below shows the Mahalanobis distances in a case where P_{init} has been underestimated. Using the equations of the Kalman filter, explain why the initial problem occurs, and also how it sort of “solves itself” naturally.



Mahalanobis distances in case of an underestimation of P_{init} .

Exercise 2 – Robot parametrization (15 min)

Wheel	L	α	d	β	γ
1	a	π	o	β_1	o
2	a	$-\pi/2$	o	β_2	o
3	b	o	c	β_3	$-\pi/2$
4	b	$\pi/2$	c	β_4	$\pi/2$

Table 2.1. Parameters of the robot.

Using the table above, give the schematic representation of the robot. The type of the wheels is not indicated but can be inferred from the table. The figure should show vectors $x_1, \dots, x_4$, the dimensions a, b, c and the angles β_i . Vectors x_i and angles β_i will have the suffix f, s, c or g as appropriate, e.g. x_{1f}, β_{2c} . For β angles, make sure they are indicated by a circular arrow with the origin at the zero position of the angle. Frame R_m must be drawn. I recommend you first draw on scrap paper and **copy a clean result** on the answer sheet once you are done. Clarity and cleanliness of the figure will be evaluation criteria.

Exercise 3: Robot modeling (30 min)

Wheel	L	α	d	β	γ
$1s$	a	o	o	β_{1s}	o
$2c$	b	$\pi/2$	c	β_{2c}	o
$3c$	b	$-\pi/2$	c	β_{3c}	o

Table 3.1. Parameters of the robot.

We want to establish the equations of the posture kinematic model of the robot of table 3.1.

- **3.1.** Give the configuration vector.
- **3.2.** Calculate matrices J_1, J_2, C_1, C_2 and C_1^* .
- **3.3.** What is the dimension of the kernel of C_1^* ?
- **3.4.** Determine a base of the kernel of C_1^* . Make sure it is a base **whatever the value** of β_{1s} .
- **3.5** Calculate the posture kinematic model of the robot.

Exercise 3: Observability (30 min)

The posture of a robot is governed by the following differential equations:

$$\begin{aligned}\dot{x} &= u_1 \cos(\theta + \beta) \\ \dot{y} &= u_1 \sin(\theta + \beta) \\ \dot{\theta} &= u_2\end{aligned}$$

In these equations, x , y and θ are the posture of the robot, and β is the steering angle of the unique steerable wheel of the robot.

The robot is equipped with a GPS sensor, the antenna of which is located at the reference point M of the robot. Remember the GPS measures the position of its antenna in the absolute reference frame.

The goal of the exercise is to study the observability of the posture of the robot.

- 3.1. Give a complete state-space definition of the system, suitable for the observability study.
- 3.2. Calculate the gradients dg_i of the components of the observation function.
- 3.3. Calculate the first order Lie derivatives Lfg_i of the components of the observation function and its gradients $dLfg_i$.
- 3.4. Find the only inputs such that the sufficient observability condition is not satisfied by using the already calculated columns of the observability matrix. **Important:** establish that the condition is not satisfied **if and only if** the value of the inputs is the one you have determined.
- 3.5. Without explicitly calculating higher order Lie derivatives, can you tell whether they will help or not, when the input is the one determined in 3.4.

Exercise 4: Path planning (15 min)

4.1. Is the extension of the Bug1 algorithm to a circular robot feasible? If yes, under which conditions and how?

Use a maximum of 10 lines of text.

4.2. What would be the main difficulties faced when trying to implement the Bug1 algorithm on a real robot?

This question can potentially be developed a lot. I expect here only the most relevant facts. Make sure you do not cover again the arguments of question 4.1.

Use a maximum of 10 lines of well chosen text.